

TAHA ARAB

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SUMMARY

Senior Mechanical Engineering Honors student with hands-on experience taking hardware from concept through fabrication, integration, and test - including a plant-adopted mechanical fix designed during an industrial internship. Proficient in SolidWorks, MATLAB, and precision prototyping. U.S. citizen eligible for security clearance.

EDUCATION

University of Houston, Cullen College of Engineering - Houston, TX *Expected May 2027*
B.S. Mechanical Engineering | Minor in Mathematics | GPA: 3.28 | Honors Program
Coursework: Finite Element Analysis, Thermodynamics, Fluid Mechanics, Solid Mechanics, Statics, Dynamics, Material Science & Lab, Experimental Methods & Lab, Numerical Analysis, Computing for Engineers (MATLAB)

SKILLS

Design & CAD: SolidWorks, Fusion 360, SketchUp, OpenSCAD - assemblies, tolerancing (GD&T), detailed drawings
Fabrication: 3D Printing (Ultimaker Cura), CNC Machining, Laser Cutting (RDWorks)
Embedded & Electronics: Arduino programming, servo/LED integration, remote-triggered actuation, circuit wiring and debugging
Laboratory: Tensile, fatigue, Charpy impact, creep, corrosion, heat treatment; pressure transducer & thermocouple calibration; velocity distribution in ducts; stress/strain measurement
Analysis & Computing: MATLAB, Tableau, SQL, R, Excel, SAP, Power Automate, JavaScript, React, Next.js

PROJECTS

CAD-Based Plane Fabrication *Jan 2025 - May 2025*

- Led all SolidWorks modeling and assembly integration for a 4-person team - fabricated components via 3D printing, CNC, and laser cutting within strict weight and clearance constraints; earned maximum score
- Owned mechanical design lead, project management, and team coordination: assigned tasks, resolved tolerance and fit issues, and verified structural integrity of the full assembly

Automated Sorting Machine (Design-Build-Test) *Jan 2025 - May 2025*

- Designed and built a fully functional remote-triggered electromechanical system - integrated servos, LEDs, and Arduino microcontroller from scratch
- Diagnosed and resolved a mid-test electrical failure under time pressure, restoring full system functionality through systematic hardware troubleshooting

OUTPOST - Browser-Based Tactical FPS (outpostfps.vercel.app) *Aug 2026 - Present*

- Built a 3D tactical shooter from scratch in JavaScript using Three.js for rendering and the Rapier physics engine for character collision and hitscan raycasting; implemented headshot hitbox detection, weapon recoil, and skeletal character animation (idle/run/shoot/death) with crossfade blending
- Engineered three tiers of enemy bot AI (Easy/Medium/Hard) with independent aim-spread, reaction time, and cover-seeking logic; procedurally generated arenas that scale layout to match size (1v1/3v3/5v5) and iteratively shipped the project across 10 tracked development milestones

Personal Engineering Portfolio (tahaarab.vercel.app) *Aug 2026 - Present*

- Designed and built a personal portfolio site (Next.js, React, Tailwind CSS) featuring an interactive AI chatbot "digital twin" that answers recruiter questions about my background and technical projects

WORK EXPERIENCE

INEOS Olefins & Polymers USA - Engineering Intern *May 2025 - Aug 2025*

- Co-designed a thrust needle bearing assembly to replace plant-wide seized ball-and-socket joints on hotbox vents - identified corrosion and heat cycling as the root cause via root cause analysis (RCA), engineered a drop-in fix, and presented to management; adopted for new hotboxes
- Built Power Automate workflow automating contract migration and a SharePoint contractor database from scratch for 1,500+ records - cut vendor search time by 80%
- Negotiated rates with an environmental compliance vendor and conducted competitive bid analysis across two vendor pairs - reduced projected contractor spend by over 30%
- Developed MRO Vendor Dashboard in Excel via SAP data tracking top 30 vendors by annual spend; diagnosed and resolved a SAP character limit bug causing recurring data entry failures
- Shadowed maintenance PM on winches, compressor motor belt replacement, and bucket lift operations - hands-on exposure to industrial mechanical systems across the facility

EXTRACURRICULAR & COMMUNITY

Young Muslims - NeighborNet Coordinator | Core Team *Jan 2022 - Present*

- Organized weekly events, retreats, and camps for 50-200 youth; collaborated with regional leadership on city-wide planning

Community Speaker *Aug 2023 - Present*

- Monthly sermons delivered to congregations of up to 800 people; YouTube channel with 15,000+ views

HONORS & AWARDS

- UH Foundation Excellence Scholarship (Jan 2026) | Cizik Scholarship (Aug 2025) | Google Data Analytics Certificate (Aug 2024) | Dean's List (Jan 2024)